

Semi-Automated Large-Scale Dataset Annotation (Object Detection)

MS Project

Brijesh Nanda Brunda Yogananda

M.S. Computer Science & Engineering · The Ohio State University · 2026

Advisor: Dr. Hari Subramoni · Committee: Dr. Rajiv Ramnath

The Problem: Annotation Bottleneck

*Building a computer vision model requires thousands of labeled images.
Creating those labels is expensive, time-consuming, error-prone and not scalable.*

2.5 min

Per Image
(manual drawing)

10K+

Images needed
before usable model

95%

Of ML project time
spent on data prep

Manual labeling is slow & expensive

A single annotator draws ~480 boxes per hour — far below the scale most CV projects require.

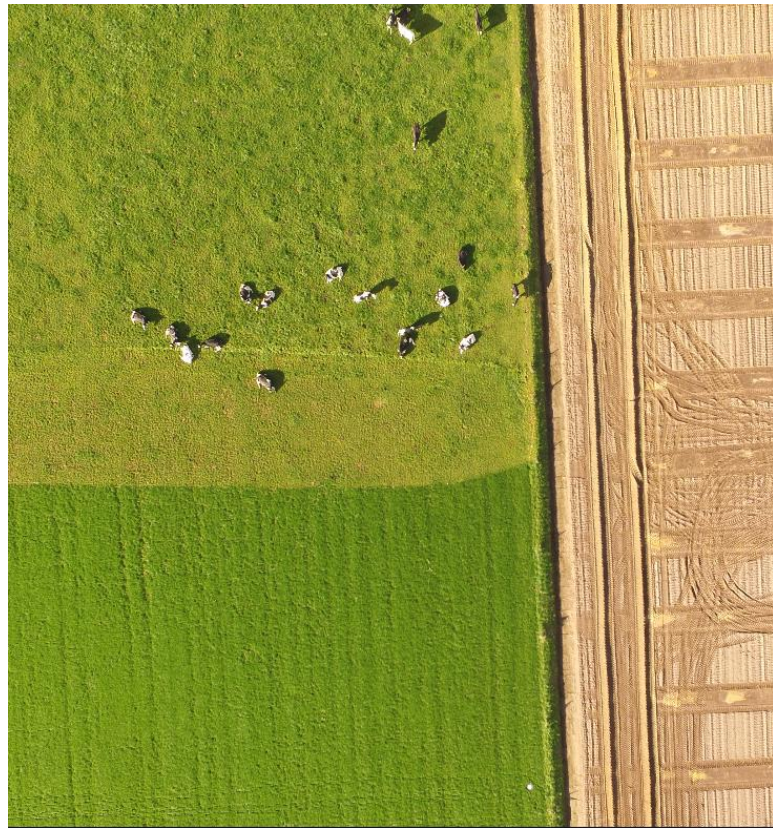
Auto-labelers produce noisy output

Fully automated systems lack domain specificity and generate false positives that still need human review.

Existing tools don't bridge the gap

Label Studio, CVAT offer UI or basic AI assist — none provide an integrated few-shot pipeline with domain-specific embedding and proposal optimization.

Sample dataset



Solution: Few-shot object detection pipeline

A two-phase pipeline combining manual precision with AI speed

PHASE 1

Manual Annotation Interface

- Bounding-box drawing
- SAM3 single-click detection
- SAM3 text-prompt detection
- NMS · confidence filtering · COCO export



PHASE 2

Few-Shot Pipeline

- Support embedding generation
- OWLv2 + SAM3 region proposers
- Cosine similarity classification
- Interactive threshold visualization and optimization

Implementation based on solution proposed in WILD-OV

Foundation Model Stack

Each model plays a distinct role — proposer or embedder

REGION PROPOSERS

OWLv2

Open-vocabulary object detector. Generates region proposals via image-conditioned patch-based queries. Also contributes patch-level embeddings.

Minderer et al., NeurIPS 2023

SAM3

Segment Anything Model 3. Provides single-click segmentation and text-prompt-driven proposal generation across images.

Carion et al., arXiv 2025

EMBEDDING MODELS

OWLv2

Contributes detection-specific visual features via its patch-level image encoder, complementing self-supervised models.

Minderer et al., NeurIPS 2023

DINOv3

Self-supervised ViT-7B. Strong semantic consistency and dense feature quality across diverse visual domains.

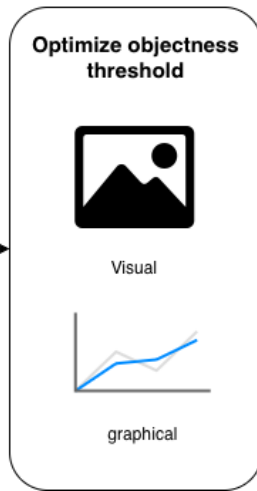
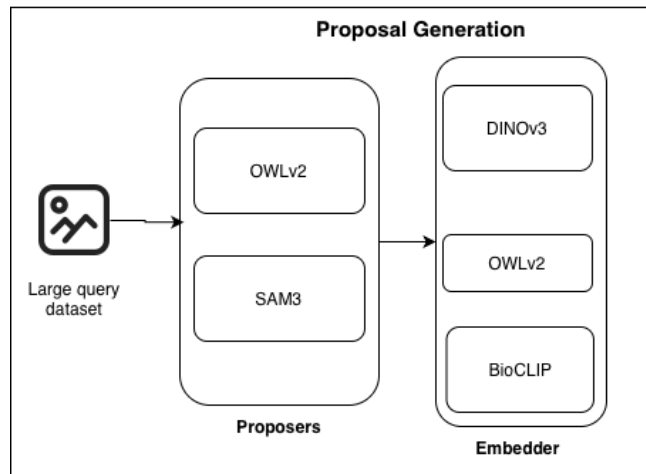
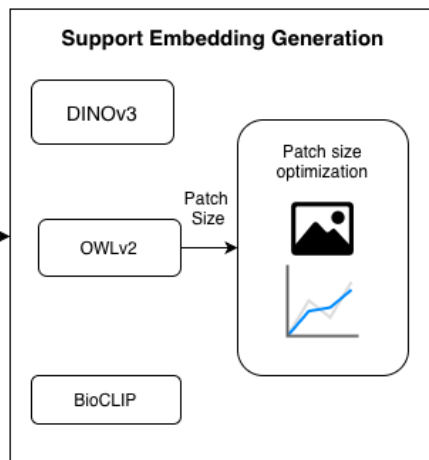
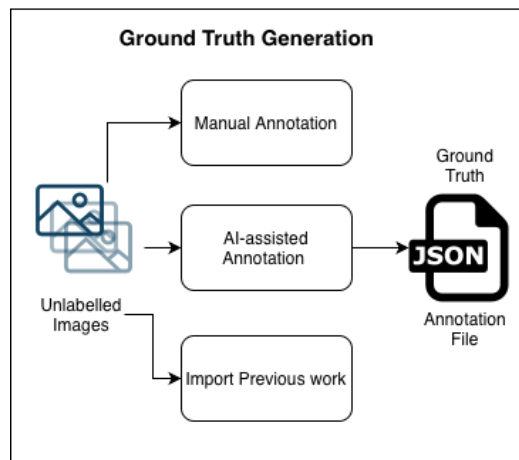
Siméoni et al., arXiv 2025

BioCLIP

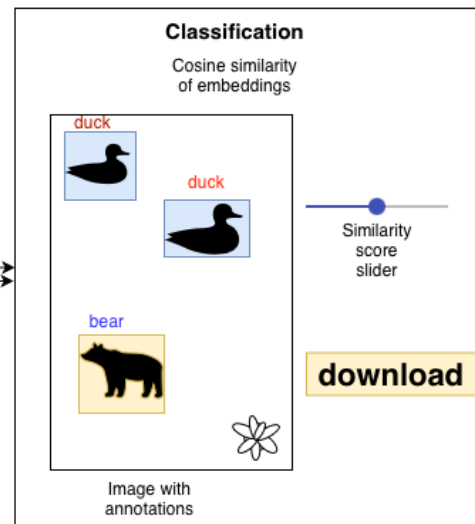
CLIP-based model specialized on biological imagery. Excels at species and organism identification.

Stevens et al., CVPR 2024

Smart Labeller



Support
Embeddings



Proposals
Embeddings

Phase 1: Manual Annotation Interface

Three annotation modalities unified in one tool



Bounding Box Drawing

Click-and-drag to create boxes. Class labels assigned from user-defined label set. Stored in COCO JSON and tool-specific format. Maximum precision for edge cases.



SAM3 Single-Click

User clicks an object — SAM3 generates a precise segmentation mask and computes the enclosing bounding box. Reduces per-instance time from 3 s to 0.4 s (85% reduction).



SAM3 Text-Prompt

Annotator types a class description (e.g. "cow") — SAM3 proposes all matching instances. Annotator filter proposals based on confidence and perform NMS for suppress overlapping boxes. Fastest mode at 0.8 s/image.

Post-processing: NMS (configurable IoU) · Confidence filtering · Import existing annotations · COCO / YOLO export

Phase 2: Few-Shot Annotation Pipeline

Four sequential stages from ground truth to annotated dataset

01

Support Embeddings

DINOv3 + BioCLIP + OWLv2
features concatenated from
GT crops

02

Region Proposals

OWLv2 patch sweep or SAM3
text prompting (visual) on
query images

03

Cosine Similarity Classification

Proposal embeddings
matched to support set via
cosine similarity

04

Review & Export

Annotator reviews auto-
labels; exports COCO JSON /
Tool specific JSON

*Ground truth
annotation set*

Patch size optimization (IoU-vs-patch-size curve) runs between stages 1 and 2 to select optimal OWLv2 inference resolution.

Custom PyTorch-Based Data Loader

Eliminating the I/O bottleneck in large-image pipelines

Sequential tiling bottleneck

- Single-process image reading
- Disk I/O blocks GPU between batches
- Intermediate tiled files written to disk
- No model-specific preprocessing



Custom DataLoader solution

- Multi-worker parallel crop extraction (PyTorch num_workers)
- In-memory patch generation — no intermediate disk writes
- Model-specific batch collation (OWLv2, DINOv3, BioCLIP, SAM3)
- Dataset interface: accepts annotation file + image list → iterable

Result: GPU stays continuously fed — substantially higher annotation throughput on large datasets.

Patch Size Optimization

Data-driven hyperparameter selection for OWLv2 proposals

The Challenge

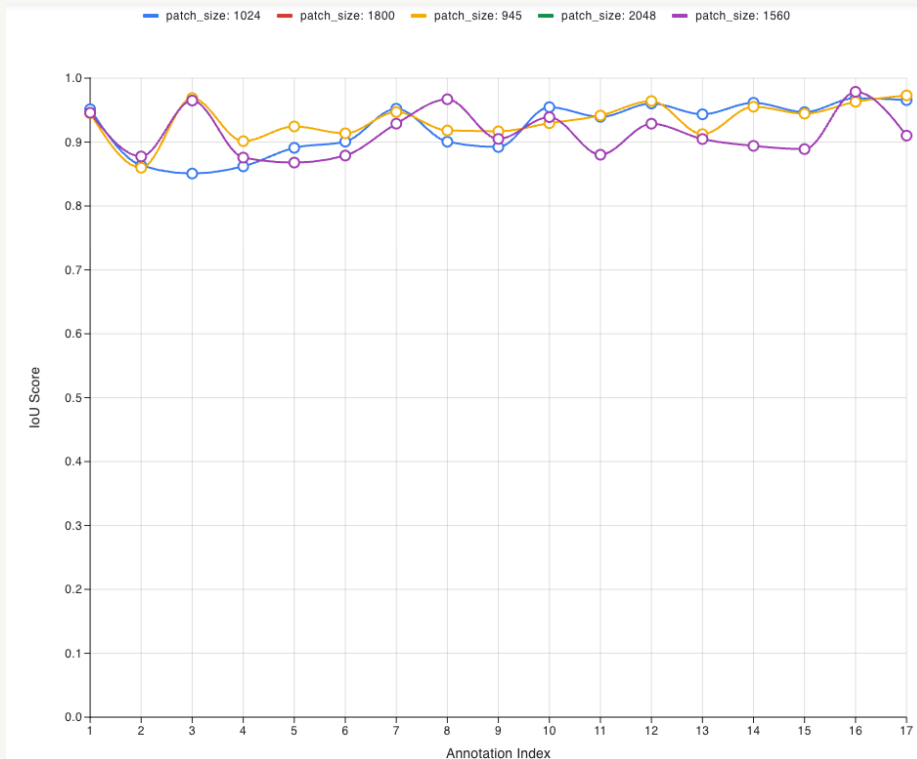
OWLv2 proposal quality is sensitive to patch size used during inference.

- Too small: misses large objects (insufficient context)
- Too large: loses precision on small instances

No principled way to choose without domain-specific evaluation.

Our Solution

For each candidate patch size, compute OWLv2 proposals on the ground truth set and measure IoU against ground truth boxes. Plot the IoU-vs-patch-size curve and select the best patch size.



Region Proposal Generation

Two complementary strategies for candidate box generation

OWLv2 — Patch-Based Detection

- Sliding-window patch sweep at the optimized patch size.
 - Open-vocabulary — no fixed class vocabulary required.
 - Objectness scores ranked; low-confidence proposals filtered.
-

OR

SAM3 — Text Prompting

- Text prompt “visual” to SAM3.
 - Detects and segments all matching object instances
 - Not constrained by class vocabulary or text descriptions.
-

Classification via Embedding Comparison

Cosine similarity matching with interactive threshold control



Above threshold → Assigned class label of nearest neighbor in support embedding space

Below threshold → Discarded as background or unrecognized instance

Interactive Confidence Threshold Visualization

Real-time slider display shows accepted/rejected annotation counts as threshold changes

Experimental Setup

Hardware · dataset · evaluation protocol

Hardware

- OSC GPU nodes — NVIDIA A100
- SDSC Expanse GPU nodes — NVIDIA V100
- SAM3 service on NRP via Kubernetes

Dataset

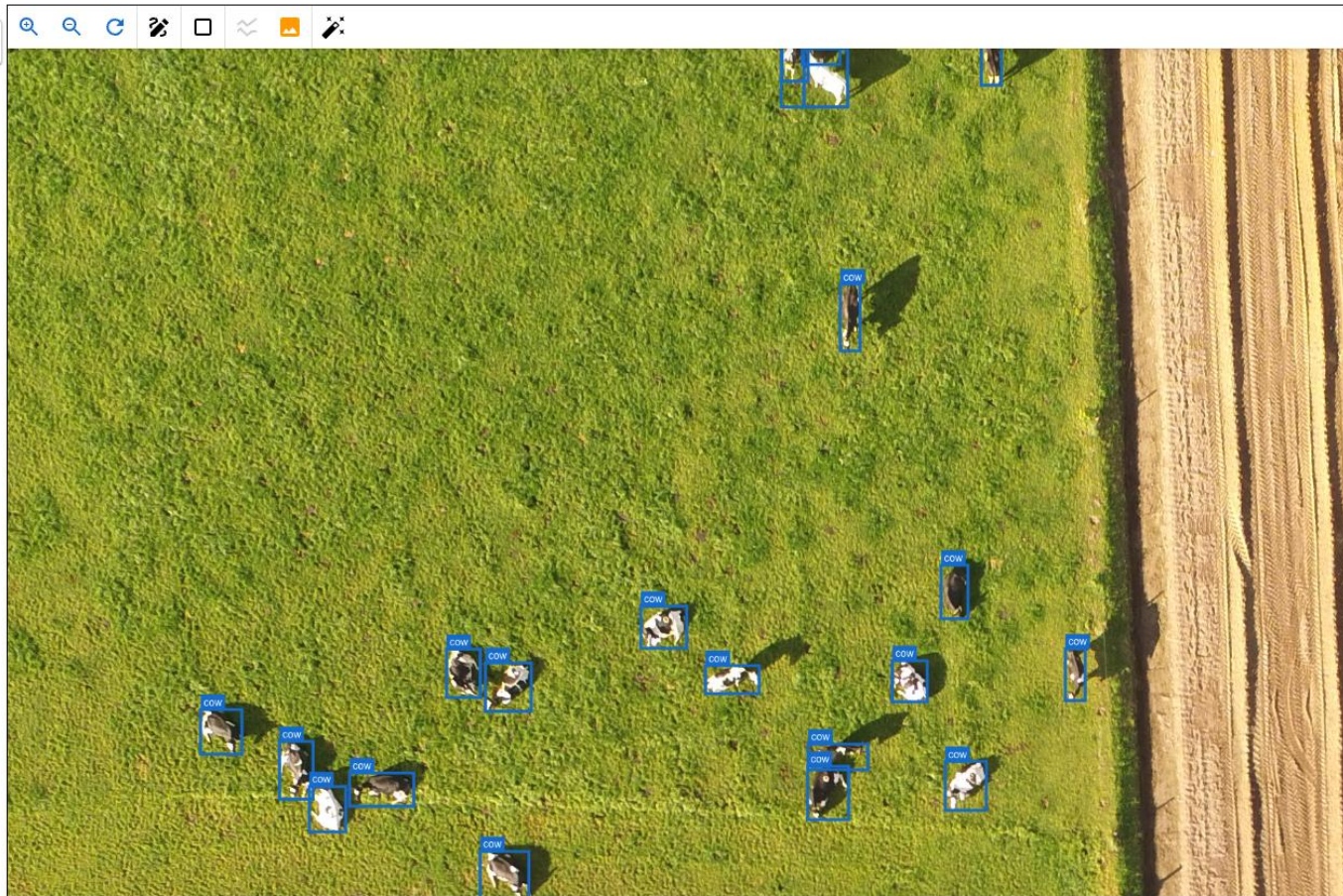
- 800 labeled images · 2 classes · 4 sets
- Drone-captured aerial imagery (WILDS data)
- Few-shot: 3 images · 42 instances

Model Checkpoints

- OWLv2: google/owlv2-large-patch14-ensemble
- DINOv3: facebook/dinov3-vitl16-pretrain-lvd1689m
- BioCLIP: imageomics/bioclclip (Pybioclclip library)
- SAM3: Hugging-Face Transformers (SAM3Processor)

Eval Protocol

- Precision, Recall, F1 score
- Patch size: 1024 px
- Timing: stopwatch, 3 annotators, 200 images



Classification Job 1

Name: Classification Job 1

Models

Proposer: OWLv2 Large Patch14

Embedder: OWLv2 Large Patch14

Thresholds

Objectness: 0.20

Similarity: 0.20

NMS IoU: 0.50

Settings

Batch Size: 4

Device: CUDA

Method: Image

SAHI Tile: 960px, Overlap: 0.20

SAHI Enabled

DOWNLOAD

Similarity Scores : 0.91

detections_owlv2_0.2_0.2_20260327_153446.json

Similarity Threshold : owl2

detections_owlv2_0.2_0.2_20260329_192703.json

Similarity Threshold : owl2



detections_owlv2_0.2_0.2_20260329_224145.j

Similarity Threshold : owl2

Results: Annotation Speed & Throughput

85%

Reduction in
per-instance time
SAM3 assisted vs manual

84%

Auto-generated boxes
requiring no
human correction

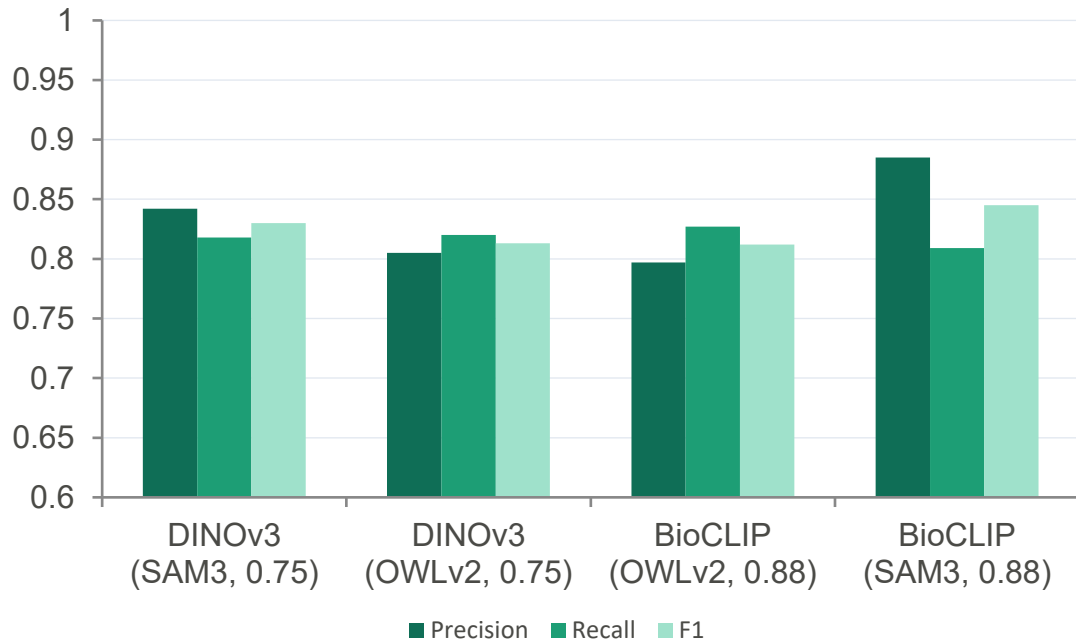
3.6 s

Per Image
few-shot auto
annotation

Mode	Avg Time/Image	Best For
Manual Bounding Box	2.5 min	All domains
Single-Click (SAM3)	5 s	Dense scenes (with contour)
Text Prompt (SAM3)	0.8 s	Common categories
Few-Shot (Auto)	3.6 secs	Domain Agnostic

Results: Embedding & Proposer Comparison

Format: embedder (proposer) at similarity threshold — 4 configurations evaluated



F1 = 0.845

Best: BioCLIP (SAM3, threshold 0.88) —
highest F1 and precision across all configs

281 FP

Fewest false positives: BioCLIP + SAM3 —
SAM3 proposals are tighter and more
selective

**SAM3 wins on
precision**

Both SAM3 configs outperform OWLv2
configs on precision (0.842 / 0.885 vs 0.805
/ 0.797)

Future Directions & Summary

Segmentation output

Extend pipeline to export instance masks alongside bounding boxes using SAM3's mask-level output

Active model selection

Auto-benchmark embedding + proposer configurations against GT set and select highest mAP configuration

Active learning

Route high-uncertainty images (low cosine similarity) to human annotators first to maximize label efficiency

Video stream ingestion

Frame extraction, temporal deduplication, and SAM3 video tracking for camera feeds and drone footage

Format expansion

YOLO TXT, Pascal VOC, DICOM-SR, GeoJSON — broader downstream framework compatibility

Training integration

Direct coupling with YOLO / MobileNet-SSD / EfficientDet — raw images to trained model in one workflow

85% faster · 40× throughput · 84% auto-correct — a practical, scalable annotation infrastructure for object detection dataset creation.

Technology Stack

Tools, infrastructure, and integrations powering the pipeline

FRONTEND

- TypeScript
- React Remix

BACKEND

- Python
- Flask API

DATABASE

- PostgreSQL

HPC INFRASTRUCTURE

- OSC (Ohio Supercomputer Center)
- SDSC (San Diego Supercomputer Center)
- NRP (National Research Platform)

CONTAINERISATION

- Docker
- Kubernetes

ICICLE INTEGRATIONS

- Patra — model card registry
- Tapis — HPC job scheduling

Releases

Smart Labeler release history

December
2025

v1.0

[Phase 1 manual annotation + OWLv2 few shot detection]

- Manual Annotation Interface
- OWLv2 based few shot annotation

March 2026

v2.0

[Patch , confidence threshold optimization + SAM3 assisted annotation]

- Patch size optimization + confidence threshold optimization
- SAM3 Assisted annotation
- Multi query file support

May 2026

v3.0

[Complete Annotator revamp + Multi proposer and embedder support]

- Fully revamped annotator (several new features)
- Multi proposer and embedder support
- Patra model card integration

Upcoming



Any Questions?

Brijesh Nanda Brunda Yogananda

Thank You

Semi-Automated Large-Scale Dataset Annotation (Object detection)

Advisor: **Dr. Hari Subramoni** · Committee: **Dr. Rajiv Ramnath**

The Ohio State University · 2026